

# SHIN-CHAN: Social Media Threat Intelligence Engine

AI-Powered Cyber Forensics & Coordinated Inauthentic Behavior (CIB) Detection for Police Cyber Cells

IBM Bob AI Innovation Hackathon · Track 2 (Cyber Forensics) · Problem Statement #06

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# The Problem: Modern Information Warfare in India

## Why Traditional Keyword Filtering and Post-by-Post Scanning Fail

- Coordinated Amplification: Botnets & sock-puppets amplify rumours (e.g. 2020 Delhi riots, 2022 flare-ups) faster than manual human review.
- The Isolation Paradox: Individual posts appear harmless in isolation (asking questions, sharing links) while coordinating offline mobilization.
- Evidence & Legal Latency: Cyber cells lack automated pipelines to map digital activity to BNS 2023 / IT Act sections before street violence erupts.
- Evidence Standards: Need for tamper-evident digital evidence chains compliant with Section 63 Bharatiya Sakshya Adhiniyam (BSA) 2023.

# The Key Insight: Behaviour First, Content Second

## Structural Multi-Signal Coordination Detection Prior to Natural Language Understanding

- Phase 1 (Behavior First): Ingest posts and construct multi-signal networks across 5 behavioral layers within 60s windows.
- Co-Tweet: Time-synchronized identical posting (same second, multiple accounts).
- Co-Similar-Tweet: Paraphrased or lexico-syntactically mutated text (Jaccard similarity  $\geq 0.8$ ).
- Co-Link & Co-Reply: Shared malicious URLs and targeted reply pile-ons against journalists.
- Phase 2 (Content Second): IBM Bob investigates flagged clusters to assess threat narrative, mobilization risk, and penal provisions.

# System Architecture: End-to-End Pipeline

## Seamless Integration of Academic Forensics, High-Performance Clustering, and IBM Bob AI

- Ingestion & Normalizer: Adapts X/Twitter IO Archives, IRA Trolls, and live streams into normalized Post models.
- Coordination Engine: QUT Digital Observatory toolkit + NetworkX Louvain community detection algorithm.
- Explainable CIB Scoring (0-100): 6 forensic features (velocity, overlap, multi-signal, freshness, burst, concentration).
- IBM Bob Headless Inference: Subprocess CLI execution (``bob run``) with Pydantic schema validation & disk caching.
- FastMCP Server & React 19 Frontend: Conversational investigation in Bob Chat and rich Cytoscape graph visualization.

# IBM Bob AI Deep Integration

## Making IBM Technology Load-Bearing Across the Entire Solution Lifecycle

- Automated Threat Classifier: Bob evaluates flagged campaigns for incitement, misinformation, harassment, or benign coordination.
- Statutory Legal Mapping: Maps evidence to BNS 2023 (353, 61, 196, 351, 356, 79) and IT Act (66D, 69A) with old IPC equivalents.
- MCP Investigation Console: FastMCP server registered in .bob/mcp.json exposes real-time forensic tools to Bob Chat.
- Executive Brief Synthesizer: Bob drafts concise operational summaries for Station House Officers under the threat-brief skill.
- Cost & Efficiency: ~0.026 BOB coins per campaign (~\$0.02); instant zero-cost retrieval from local cache.

# System Evaluation & Benchmark Results

## Empirically Validated Against Synthetic Scenarios and Real-World Datasets

- 100.0% Detection Recall: Discovered and isolated all planted coordinated rings (Ring A, B, C) with zero missed malicious accounts.
- Zero False Positives: Decoy D (benign celebratory cricket chatter) scored 0/100, proving volume spikes do not trigger false alerts.
- BNS Legal Adherence: 100% adherence to authorized BNS/IT Act offenses; zero hallucinations; banned sections (ITA-66A) strictly blocked.
- Rapid Forensics: Graph construction, multi-signal edge resolution, and Louvain clustering complete in under 5s on 5,000 posts.
- Real-Data Sanity: Successfully processed and clustered slices of Russian IRA trolls and X Information Operations archives.

# Operational Value for Police Cyber Cells

## From Raw Social Streams to Court-Ready Evidence in Minutes

- Rapid Escalation: Immediate operational directives (URGENT / ALERT / MONITOR) tailored for District Control Rooms.
- BSA Section 63 Compliant Briefs: One-click print-ready HTML/PDF threat briefs with SHA-256 evidence integrity chains.
- Interactive Visual Forensics: Cytoscape network graph enables officers to explore node degrees, bridges, and ring topologies.
- Responsible AI Guardrails: Behavioral focus avoids political bias; strict non-profiling rule bans demographic/religious labeling.
- Single Command Deployment: `python src/main.py` starts the unified backend and served React frontend on port 8000.

# Team Shin-chan & Hackathon Summary

IBM Bob AI Innovation Hackathon (INNOVAI) · Track 2: Cyber Forensics

- Tirth Chokshi (Lead): System Architecture, Coordination Engine, Pipeline & Validation.
- Milind Pawar: React 19 UI, Cytoscape Network Graph, Recharts Timeline.
- Jainik Devada: IBM Bob Integration, FastMCP Server, BNS 2023 Legal Mapping.
- Jigar Jariwala: Scenario Synthesizer, Real-World Data Adapters, Evaluation Benchmark.
- Repository: <https://github.com/Tirth-chokshi/bob-ai-hackathon-shin-chan>